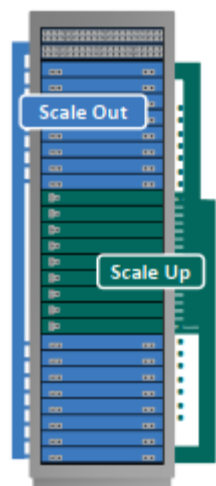


Connectivity and Memory Innovation for Large Model Era

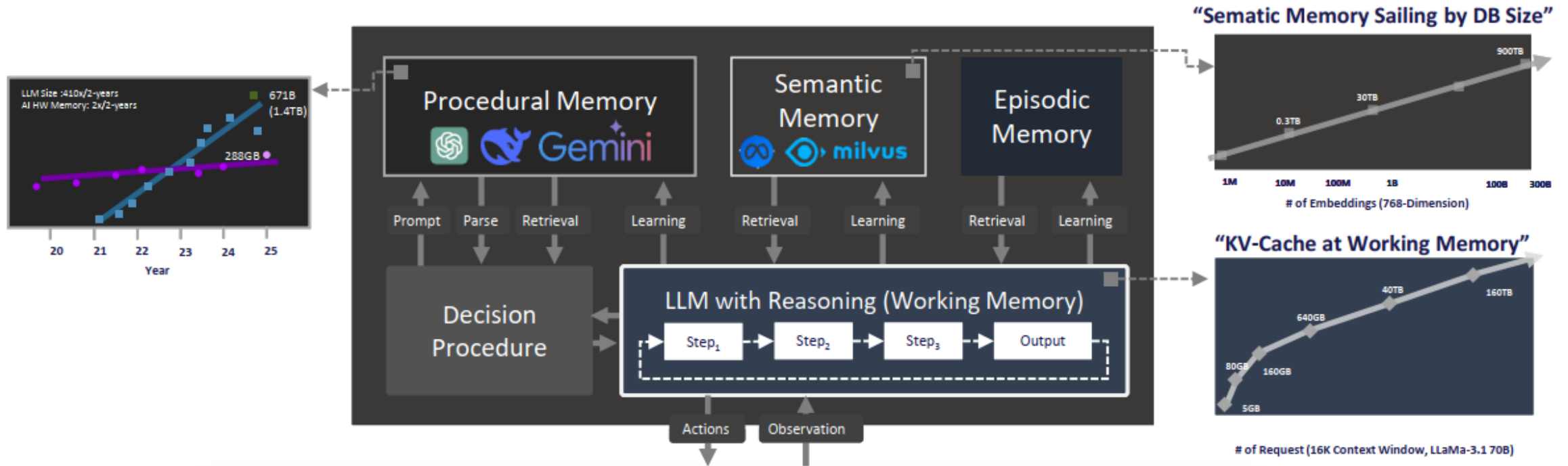


zettEngine: Economizing LLM Inference with a RISC-V CXL Memory Systems

Zettai LLC



LLM based Agentic AI: Memory View



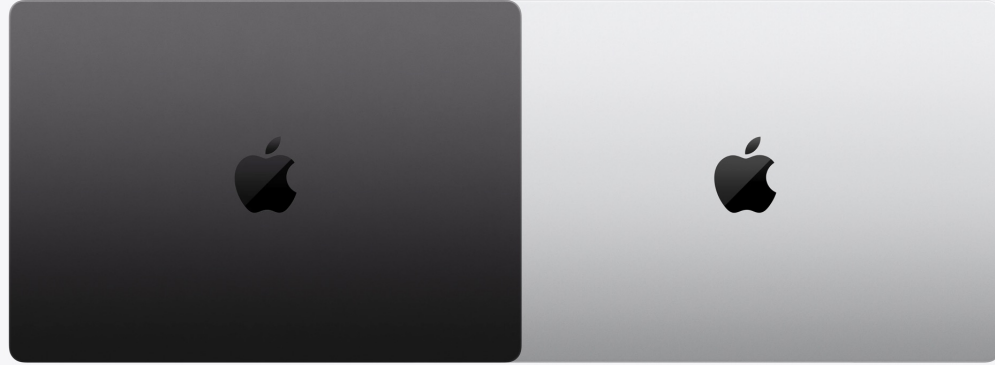
Agentic AI based on LLM with KV cache requires significant memory capacity and bandwidth relying on external databases

- (Working) A space where agent temporarily stores and processes information current tasks or reasoning.
- (Procedural) Implicit knowledge stored in LLM weights.
- (Semantic) Agent's knowledge about the world and itself.
- (Episodic) Experiences from earlier decisions (e.g. training input-output pairs, history event flows, and game trajectories)

Scalability Problem

Target Market

\$7,349.00 or \$612.41/mo. for 12 mo.*



Price

USD \$3,000



FLAGSHIP ZettEngine

Our flagship AI acceleration engine featuring direct GPU-to-CXL FPGA loading with AMX optimization. Running ktransformer with unprecedented efficiency, ZettEngine delivers 10x cheaper inference while maintaining industry-leading performance metrics.

CXL 3.0

FPGA Direct Load

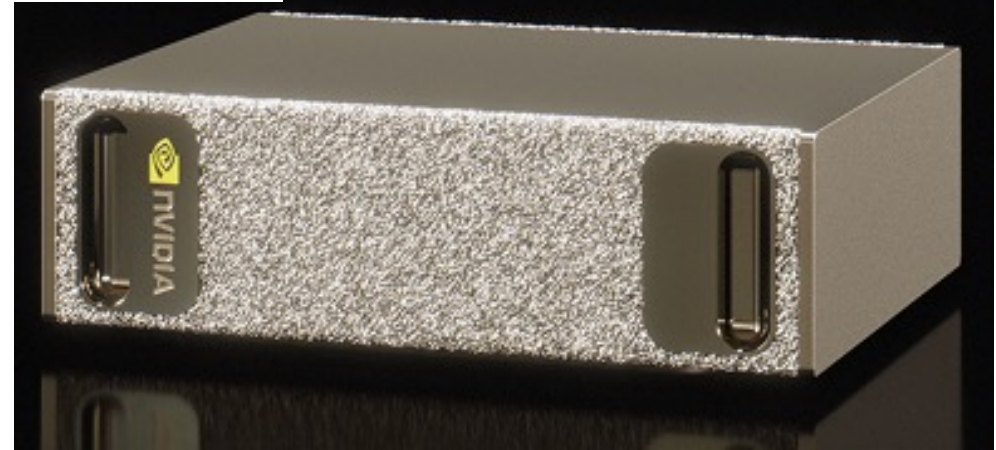
AMX Optimized

ktransformer

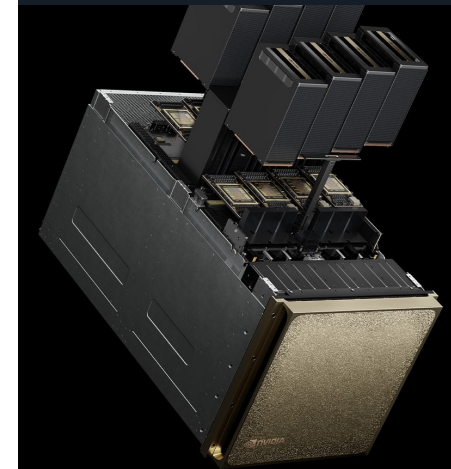
GPU Offloading

🔥 OpenClaw Native

\$4,699.00



\$45,000–\$50,000



Background and Motivation

The challenges to resolve the memory capacity and performance of Next-Gen Memory-Intensive App

1. The HBM availability, supply stability, and high TCO

Those factors are slowing down the AI server development and grow. In this new era, we should explore how to use fewer HBM. Instead, increase host memory capacity and performance. This will keep scalability, supply availability and lower TCO.

2. Memory subsystem scalability and lower the TCO requirement

The diversity of Memory-Intensive App Server architectures depends on their use cases and workloads. How to avoid the under-provisioning, over-provisioning of the memory subsystem and stranded memory are the challenges to Next-Gen Memory-Intensive App Server architectures today

3. The need for bigger memory capacity and higher performance per CPU-socket

Scaling a host memory capacity and performance per CPU socket is becoming increasingly challenging each year, yet DLRM, IMDB, AI and HPC workloads continue to drive the demand of CPU sockets toward bigger capacity and higher performance

* Host Memory (CPU-manage memory > CPU near memory, far memory, pooled memory) JBOM

Memory Tiering/Pooling Based on PCIe/CXL

Memory Tiering/Pooling can significantly enhance KVCache performance in LLM inference, especially in high-concurrency, long-context workloads

Memory Domain	GPU HBM	CPU Memory	Tiered Memory
Latency	Lowest	Low	Comparable to CPU memory
Capacity	Limited	Moderate	Scalable (TB-level)
Cost per GB	High	Moderate	Lower than GPU DARAM
KVCache Suitability	Best for host cache	Good for mid-tier	Ideal for cold/pooled/shared
Scalability	Restricted	CPU Socked-bound	Fabric-wide

Scalability > Avoid vendor lock-in, memory stranded, over provisioning

> Reduce TCO

> procurement and deployment > on demand, heterogeneous memory architecture

Memory Tiering/Pooling Based on PCIe/CXL

JBOM Use Case



Fabric management
+ CXL switch
+ CXL Memory

Host Adapter

Compatible with various host models and ensuring intact data transmission.



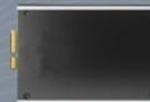
CDFP Cable & Connectors

High speed signal connection within a rack.



Flexible Memory Expansion

Up to 20 (x8) memory modules with up to X.0TB memory capacity.



Chassis Retimer

Improves signal quality and system compatibility for reliable operation.

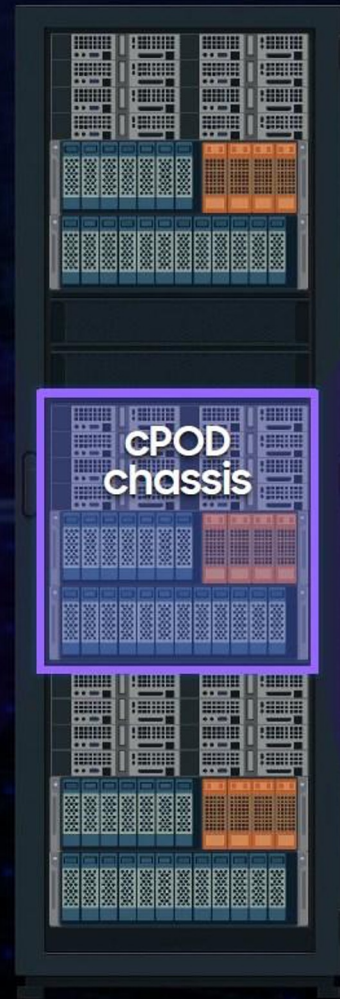
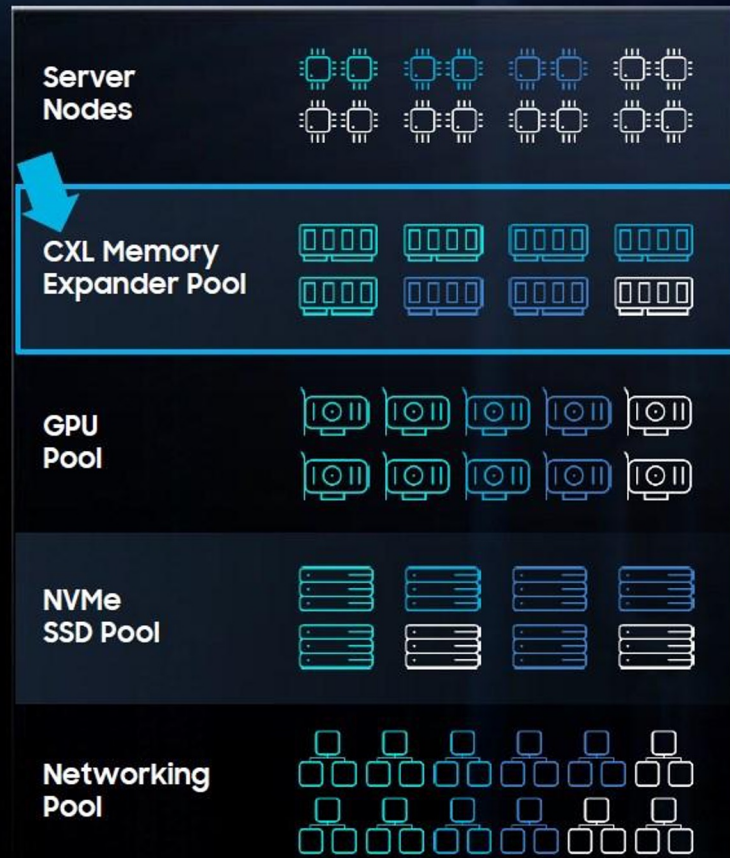


Next-Gen Memory & Interconnect Subsystem-Composable, Tiered/Scalable

avoid stranded memory increase energy efficiency reduce TCO (CAPEX,OPEX)

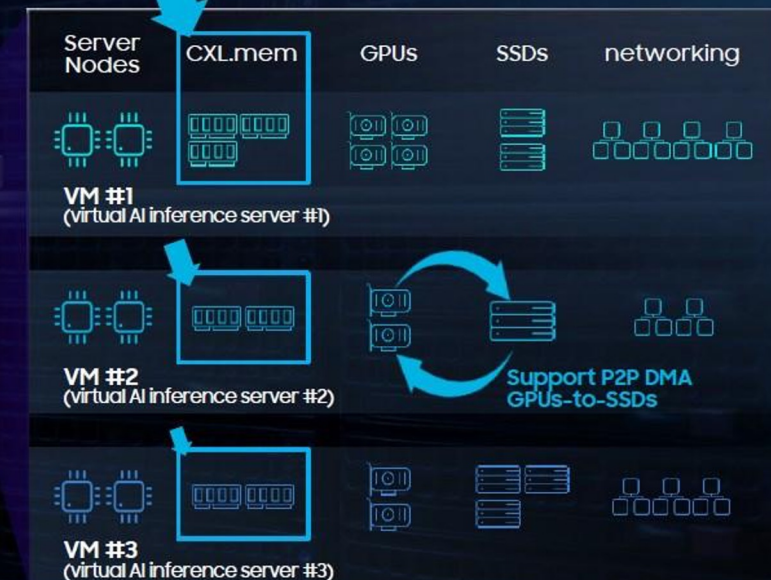
CXL Environment

Composable pod components

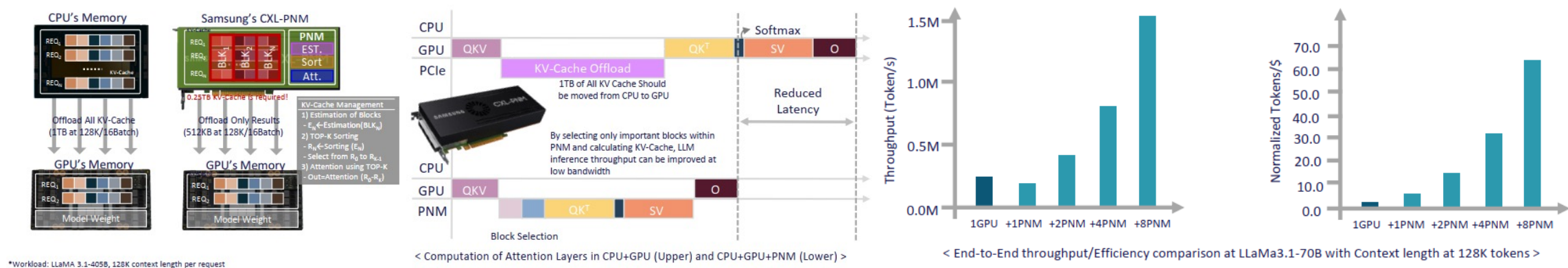


Software-Defined Virtual AI inference servers

VMs on demand



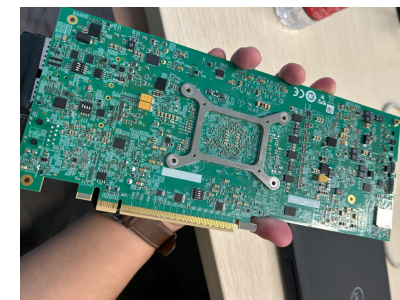
Use Case: CXL-RPU(Remote Processing Unit) A LLM-turbo



Efficient KV-Cache management running RPU for scalable LLM inference to long contexts

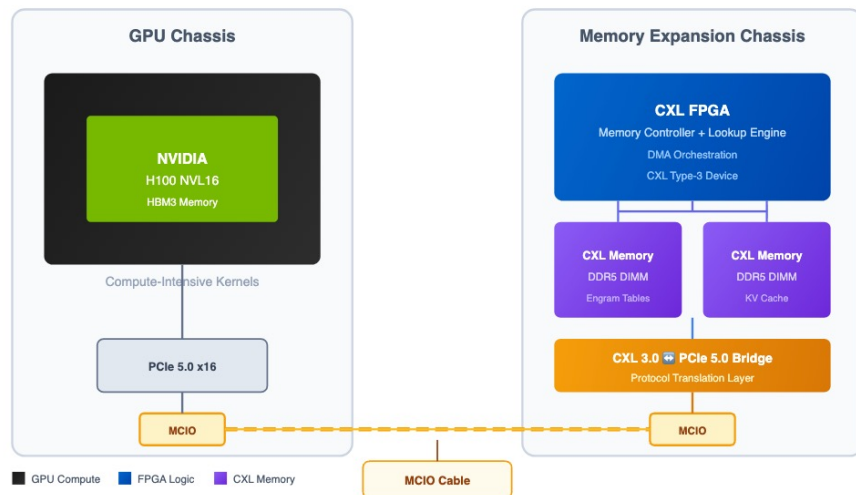
- CXL-RPU performs token block based selection directly within CXL memory module, eliminating GPU's KV-Cache offload cost
- By storing KV-Cache in CXL memory and offloading selection to RPU, GPU memory pressure and support larger batch sizes
- With the integration of DPU-like(RPU) technology, future KV Cache operations can be processed with greater speed and efficiency

CXL-RPU Altera Agilex 7 I
series/AMD Versal
Premium 2 FPGA Module

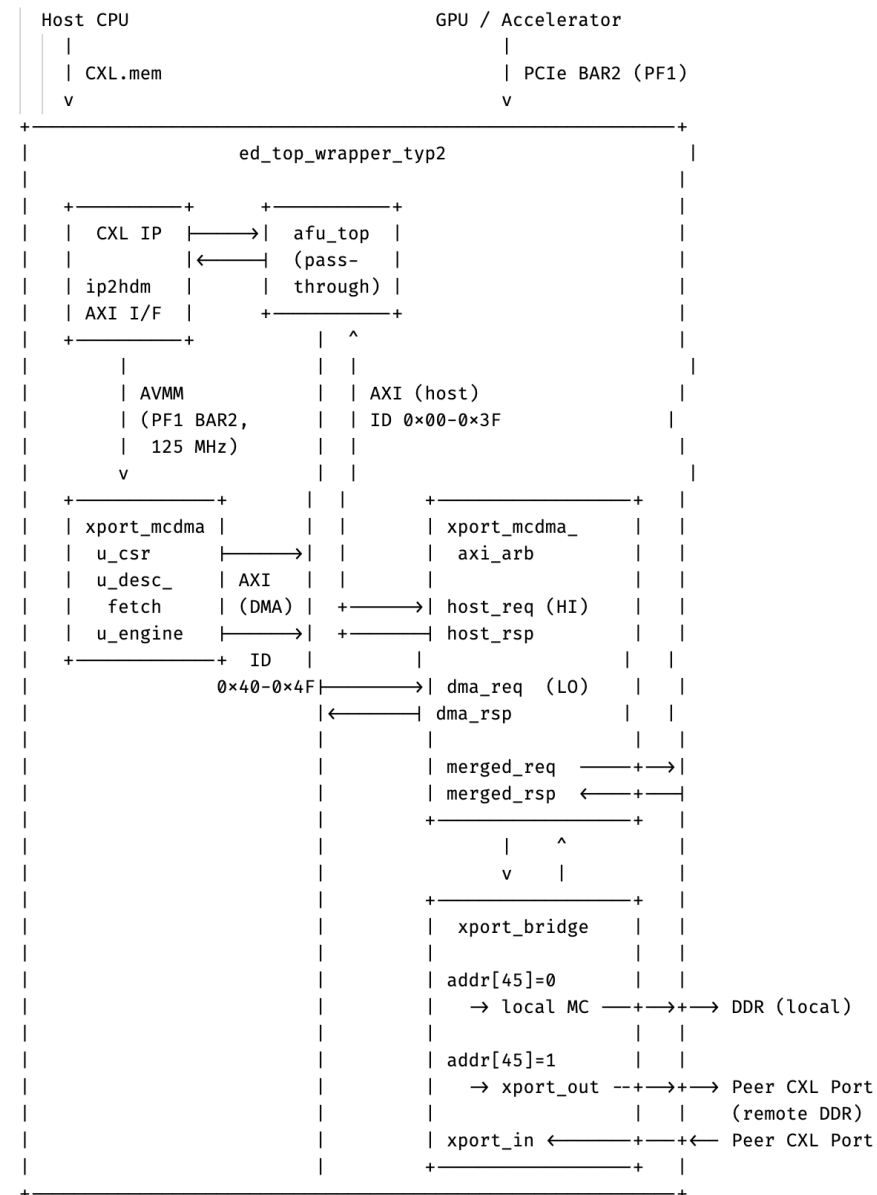


Use Case: CXL-RPU(Remote Processing Unit) A Connector

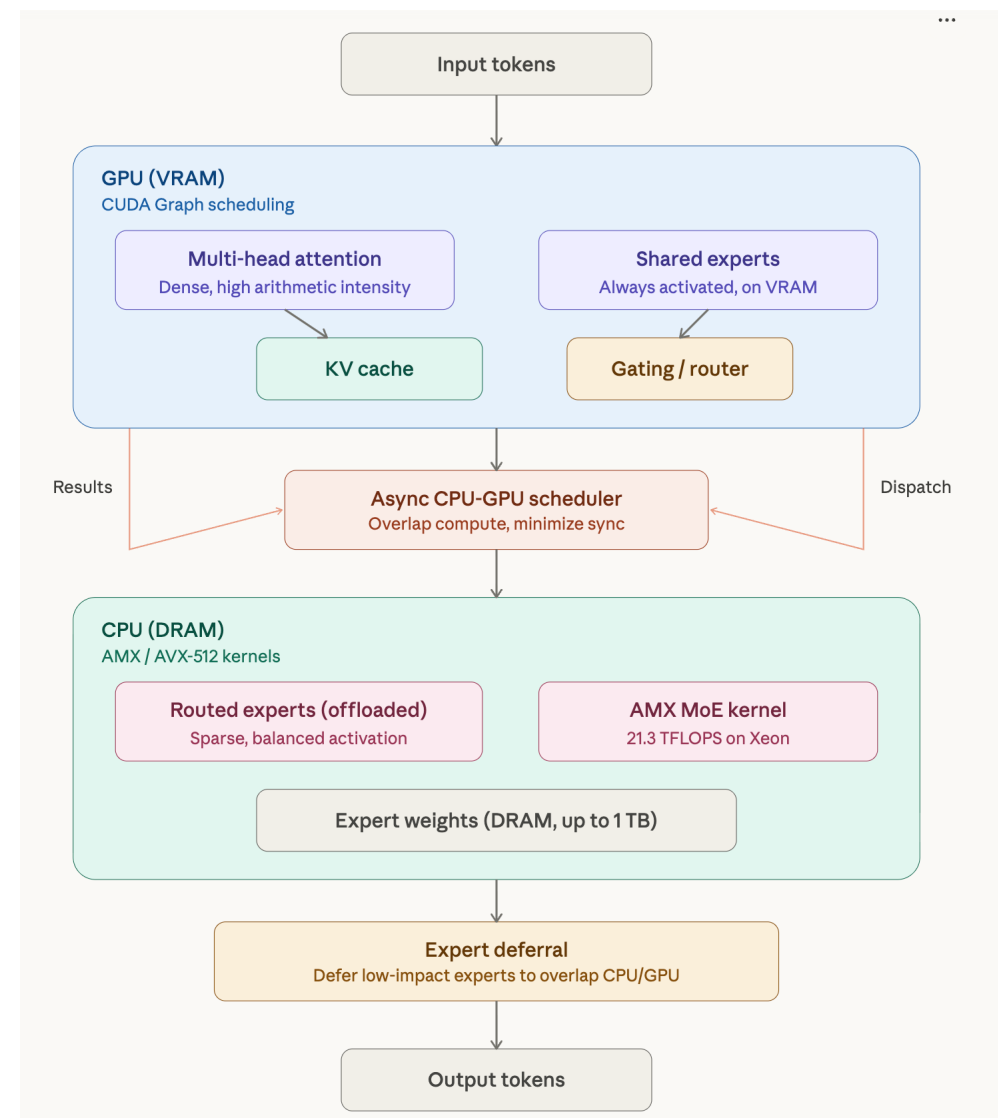
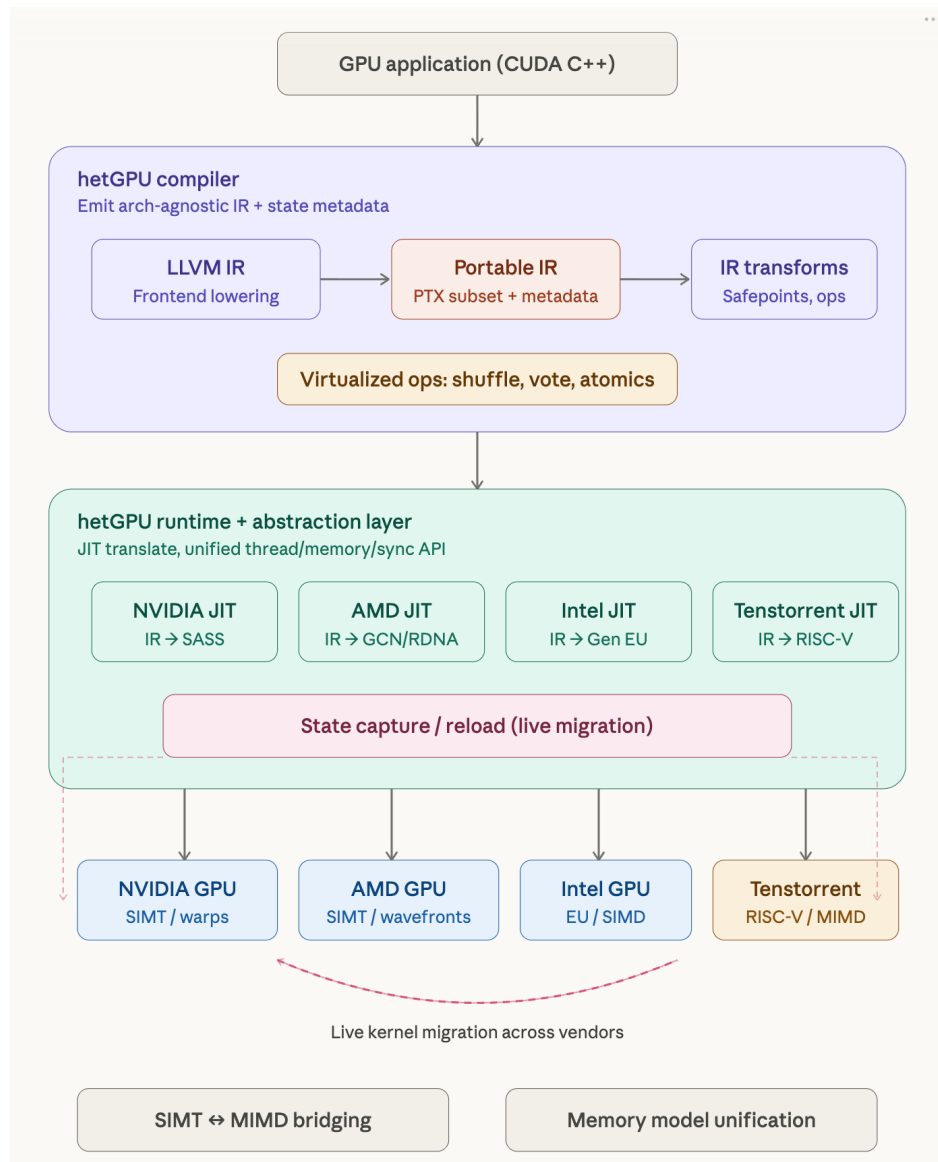
ZettEngine – Dual Chassis Architecture



- Could act as both device and host
- RISC-V Soft Core for Data Prefetching from PCIe to CXL
- Engram-style conditional memory introduces a lookup--compute separation, where large memory tables are accessed via indexed lookup with constant-time complexity
- Disaggregating HBM to CXL Memory Just like Storage Service brought by Nvidia.

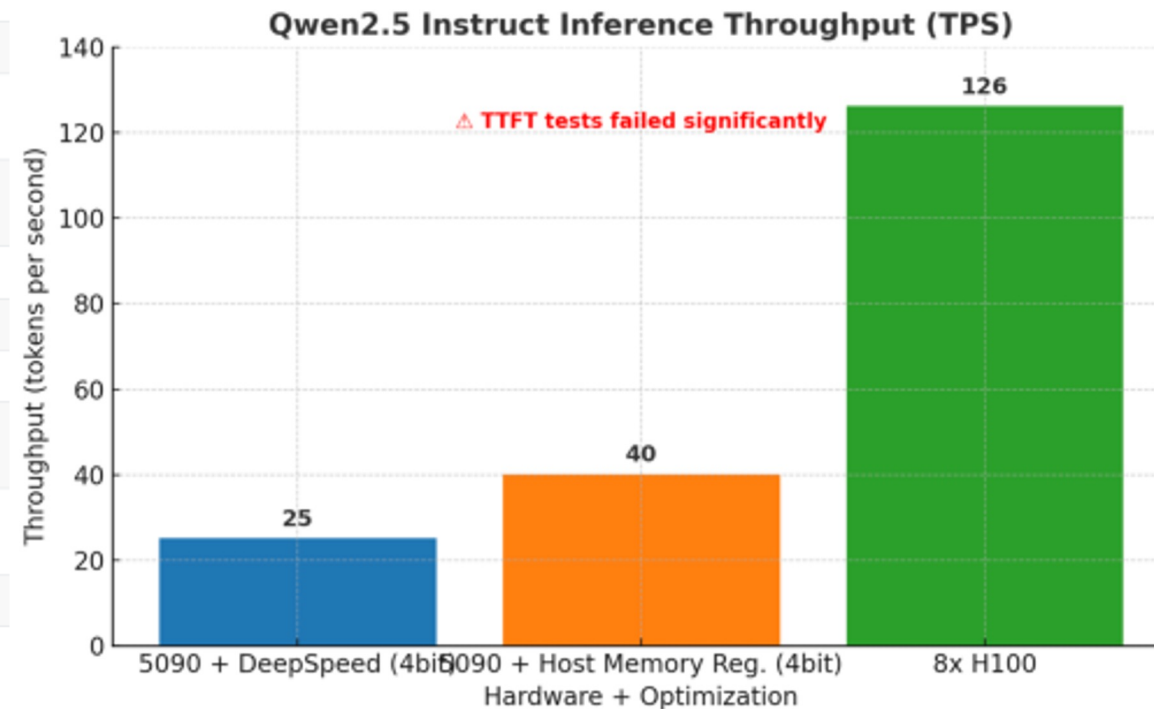


Use Case: CXL-RPU(Remote Processing Unit) A Connector



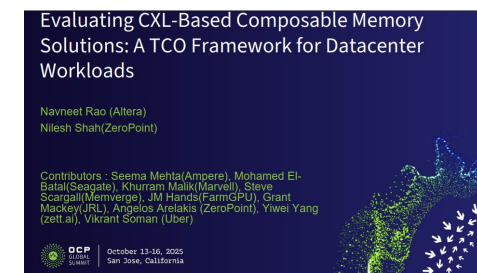
Use Case: CXL GPU Direct-Memory for lower DC-TCO

Metric	8x H100 GPUs	5090+CXL Solution	Cloud Solution
Hardware Capability	✅ Just fit run 671B (640GB < 700GB needed)	✅ Can run quantized version	✅ Full model support
Actual Performance	40 TPS	6-8 TPS (needs optimization to 40)	40+ TPS
Initial Investment	\$400,000	\$6,000	\$0
Power Cost (3yr)	\$62,100	\$15,000	Included
Hosting/Colocation (3yr)	\$108,000	\$36,000	Included
Maintenance/Support (3yr)	\$60,000	\$10,000	Included
Total TCO (3yr)	\$630,100	\$67,000	\$394,200
Cost per 1K Tokens	\$0.002	\$0.0001	\$0.003



1. Host CPU that supports CXL p2p like latest thread ripper
2. Nvidia GPU like 5090 that direct load store to memory pool
3. Altera CXL FPGA for compression and translating PCIe to CXL p2p
4. CXL memory pool with up to 1TB
5. Deepseek 671B 40TPS can be comparable to 8 cards H100

Price @15k vs. 8 cards H100 @ 400k, beat in TPS/\$, failed in TTFT, will fix through sw



OCP work

Product Profolio: PCIe/CXL based technologies

Retimer

Adapter

CDFP

CXL-MXP

CXL-PNM

CXL-Bridge

CXL-JBOM

CXL-Server

CXL-PoD

Fabric

SW

CXL MEMORY POOLING SUPPLY CHAIN

Non-exhaustive schematic



CXL Compliance and Testing



CXL Fabric Management



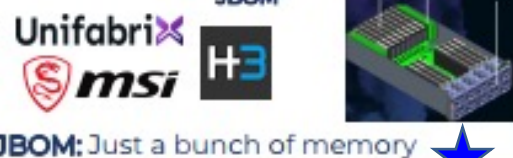
Software



Potential player



CXL Compliance and Testing



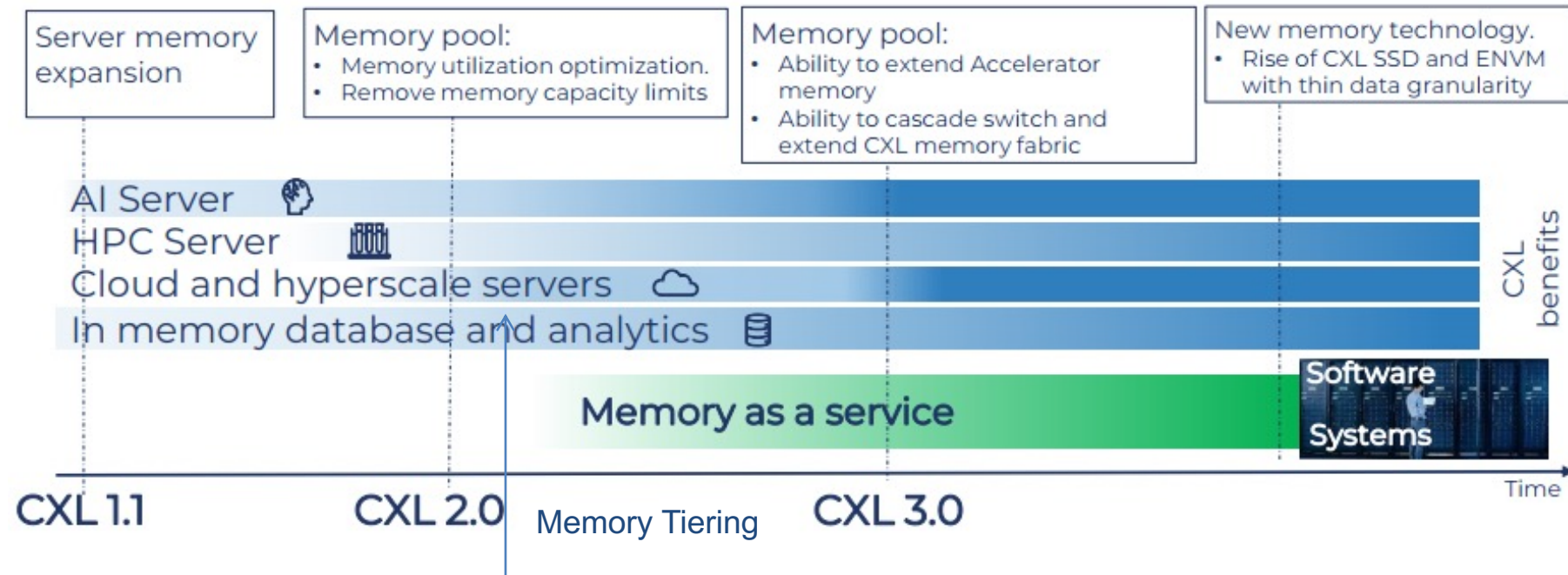
CXL Memory Expander



Memory-Processor Interface 2023 - Focus on CXL | Report | www.yolegroup.com

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Adoption Milestone



- CXL has the potential to revolutionize memory utilization and access, leading to widespread adoption as its benefits become evident to every player.
- This could be compared to the shift that occurred in the 90s to the 2000s when storage transitioned from being a CPU-attached element to complete disaggregation over networks.
- Similarly, we can anticipate the disaggregation of memory, paving the way for a new industry centered around CXL memory fabric software, systems, and services.

Monetization

- Inference-as-a-Service (IaaS) with TCO arbitrage.
- FPGA IP licensing + reference design.
- CXL memory fabric software stack.
- **Turnkey appliance for on-prem LLM inference.**
- **Developer tools and benchmarking services.**
- CXL-native CMX alternative for neoclouds.

Thank you

